

DIGITALSPY

Predictive Cyber Defence

Internal Research Paper — Landscape, Gap, Mathematics, Architecture & Rationale

NTRO / Smart India Hackathon 2026
Internal team use • Not for publication

1. Purpose of this document

This is the team’s long-form research record for understanding DigitalSpy before, during, and after implementation. It is intentionally broader than the SIH pitch deck and more explanatory than the implementation specification. It records the problem definition, what exists, what does not, why the gap matters, the mathematical framing, design alternatives considered, research risks, dataset strategy, and the reasoning behind the scoped prototype.

Core mental model: Observe → Represent → Learn Dynamics → Forecast → Explain → Contextualize → Support a Human Decision

2. The problem in one sentence

Critical-infrastructure defenders can detect suspicious traffic, correlate alerts, and analyse potential attack paths, but as an attack evolves through time they still need a reliable estimate of what the observed network state is likely to become next; DigitalSpy studies whether temporal network-state modelling can provide that forecast with probability, evidence, and measurable lead time.

3. The core distinction: detection is not forecasting

Capability	Primary question	What it sees	DigitalSpy relationship
Detection / IDS	Is something suspicious happening?	Current observations	Baseline capability
Classification	What type of malicious event is this?	Current event/window	Baseline target
Correlation	Which events form one incident?	Past + present events	Context layer
Attack-path analysis	What paths could an attacker take?	Topology, exposure, permissions, vulnerabilities	Different from temporal observed trajectory
Forecasting	What is likely to happen next?	Observed history + state dynamics	Primary research target

The distinction matters because a system can correctly detect a malicious flow yet still fail to forecast future progression. Conversely, an attack-path graph may show that a route is theoretically possible without saying that an observed adversary is likely to use it next.

4. Who is affected?

- SOC analysts facing high-volume, noisy alert streams.
- Incident responders deciding whether an evolving event is escalating.
- Network defenders in critical or restricted environments where offline analytics may be valuable.
- Security engineers who need reproducible predictive signals rather than narrative-only AI output.

5. What exists today — and where the boundary is

The team's earlier technical gap analysis reviewed commercial platforms and research across detection, correlation, exposure management, attack-path analysis, temporal modelling, and forecasting. The important lesson is to avoid a “nobody predicts attacks” claim. Existing products and research make prediction-adjacent claims, and recent work demonstrates explicit probabilistic cyber-attack regime forecasting. The defensible DigitalSpy position is therefore about the specific experimental design: a reproducible offline feature-vector state model, an explicit static→Markov→attention-LSTM progression, multi-horizon forecasting, forecast-lead-time evaluation, and an operator layer that is separated from numerical prediction.

5.1 Commercial / operational landscape

System family	Strength	Boundary relevant to DigitalSpy
Darktrace	Behavioural baselines, detection, incident correlation, probable next steps/escalation research	Strong adjacent capability; DigitalSpy must not claim commercial platforms do zero forecasting.
Microsoft security / exposure graph	Cross-source security telemetry, attack-path and exposure analysis, disruption	Strong structural/context analysis; differs from the proposed controlled temporal forecasting experiment.
ExtraHop / Vectra	Network detection, entity behaviour, lateral-movement/attack-chain analytics	Primarily operational detection/investigation; predictive framing must be defined precisely.
Tenable / XM Cyber	Exposure management and attack-path analysis	Answers where an attacker could move under environment assumptions, not the same as forecasting an observed temporal trajectory.
MITRE CyGraph / CICAT	Attack-path / incident-analysis decision support	Useful semantic/graph context; not the Week-1 numerical forecast model.
Research systems	Temporal graphs, switching state-space models, next-victim prediction	Prove that forecasting is a research area already; novelty must be narrow and evidence-based.

5.2 The gap we can honestly test

A useful engineering gap is not “future attacks are impossible to predict today.” It is: can a small, reproducible, offline system built from network-traffic feature vectors demonstrate a measurable advantage in multi-horizon future-risk/tactic forecasting, especially in forecast lead time, when compared with a static classifier and a transparent first-order Markov transition model?

USP vs gap: Gap describes what is difficult or missing; USP describes what DigitalSpy specifically does. The USP is the end-to-end, offline, explainable, lead-time-evaluated forecast pipeline — not the mere existence of a forecasting gap.

6. Research question and hypothesis

Primary research question:

Can temporal modelling of observed network states provide calibrated multi-horizon forecasts of future malicious behaviour with useful lead time, compared with static detection and a simple Markov transition model?

Primary hypothesis H1:

Attention-LSTM temporal dynamics will achieve better horizon-averaged tactic macro-F1 than the first-order Markov baseline, and will provide positive forecast lead time on a meaningful fraction of attack instances.

Secondary hypothesis H2:

Adding temporal history will improve the predictive task relative to a static Logistic Regression baseline under a chronological evaluation protocol.

Null result is acceptable and scientifically informative: if the hypotheses fail, the team should identify whether the state representation, labels, horizon, or model is the limiting factor.

7. Is the problem mathematically definable?

Yes, provided we formulate it as probabilistic conditional forecasting rather than deterministic future knowledge.

O_t = observed network telemetry at time t
 $S_t = g(O_{\{window\ t\}}) \in \mathbb{R}^{24}$
 $H_t = (S_{\{t-h+1\}}, ..., S_t)$
Goal: estimate $P(Y_{\{t+1:t+K\}} \mid H_t)$ and, in the stronger state form, $P(S_{\{t+1:t+K\}} \mid H_t)$

The prototype uses a feature state S_t as the observed representation. It does not claim that S_t contains the full hidden cyber state, attacker intent, or every causal variable. Therefore the model is best interpreted as estimating probabilities over plausible future behavioural outcomes conditional on observed history.

7.1 Why exact future prediction is not the goal

At time t , several futures may be consistent with the same observations: the activity may continue, stop, become benign, escalate, or switch behaviour. A deterministic claim would be brittle and scientifically unjustified. A probability distribution is a natural representation of uncertainty.

8. S_t versus Z_t — two layers of state

The final specification intentionally separates the rich continuous state used by Tier 3 from the compact discrete state used by Tier 2.

Symbol	Definition	Role
S_t	24-dimensional normalized continuous feature vector	Input to Attention-LSTM; preserves detailed traffic information
Z_t	Six-state discrete behavioural abstraction	Input/output of Markov baseline; gives a transparent 6×6 transition matrix
H_t	20-step history of S_t	Context for the LSTM forecast
$Y_{\{t+k\}}$	Future risk or tactic target	Supervised forecast outputs

9. Exact S_t feature schema (24 features)

Group	Features	Count
Volume / timing	flow_count; total_bytes; total_packets;	6

	mean_flow_duration; bytes_per_sec; packets_per_sec	
TCP flag ratios	syn_ratio; ack_ratio; fin_ratio; rst_ratio; psh_ratio	5
Inter-arrival timing	mean_iat; std_iat; max_iat	3
Packet size	mean_packet_len; std_packet_len	2
Diversity	distinct_dst_ips; distinct_dst_ports; distinct_src_ports_used	3
Directionality	fwd_bwd_byte_ratio; mean_init_win_bytes_fwd	2
Packet-derived	ttl_variance; tcp_window_std; fragmentation_ratio	3

All 24 values are computed per source host, per 10-second window. Flow-level values are aggregated from pre-computed CICFlowMeter rows rather than rebuilding every original feature. Packet-derived fields are extracted from a small PCAP sample and fused into the same schema. Normalization statistics are learned from the training period only.

10. Exact label-precedence rule

A host-window may contain multiple raw attack labels. The project deliberately resolves this ambiguity with a deterministic precedence rule: later-stage activity is treated as the more urgent and informative signal, while earlier-stage activity co-occurring in the same window is treated as background.

IMPACT > C2 > LATERAL_MOVEMENT > INITIAL_ACCESS > RECON > BENIGN

Raw CIC-IDS2017 label	Bucket	Rank
DoS Hulk / GoldenEye / slowloris / Slowhttptest / DDoS	IMPACT	1
Bot	C2	2
Infiltration	LATERAL_MOVEMENT	3
Web Attack (Brute Force/XSS/SQLi), FTP-Patator, SSH-Patator, Heartbleed	INITIAL_ACCESS	4
PortScan	RECON	5
BENIGN	BENIGN	6

tactic(window) = min(rank(label)) over every label present in the host-window; BENIGN only if no attack label appears.

Important: These six buckets are a project-defined modelling taxonomy mapped from raw dataset labels. They are not a claim that CIC-IDS2017 supplies native ATT&CK tactic ground truth.

11. Temporal design

The locked prototype uses 10-second fixed non-overlapping windows per source host. Windows with fewer than two flows are omitted. The LSTM receives $h = 20$ windows, representing 200 seconds of observed context.

$H_t = [S_{t-19}, S_{t-18}, \dots, S_t]$

Forecast horizon $K = 5$; each step is one 10-second window, giving a nominal 50-second look-ahead.

11.1 Why these numbers?

They are explicit engineering choices, not universal constants. Ten seconds provides temporal resolution for bursty scanning and session behaviour; 20 windows provide several minutes of history without creating an unnecessarily large sequence. $K=5$ creates a visible degradation curve and a genuinely future-oriented forecast without demanding long-range simulation.

The Day-1 dataset audit may validate or challenge the choice, but after the validation design is frozen the team should not repeatedly alter window size or horizon in response to test performance.

12. Dataset strategy and verification

CIC-IDS2017 is the primary Week-1 benchmark because the official dataset documentation provides labelled flows, PCAPs, a five-day capture, attack schedules, and CICFlowMeter-derived traffic features. The file contents in the local copy should be treated as the final source of truth for the experiment.

12.1 Verify the actual files before splitting

```
import pandas as pd, glob
for f in sorted(glob.glob('cic2017/*.csv')):
    df = pd.read_csv(f, usecols=lambda c: c.strip() == 'Label')
    print(f, df.iloc[:,0].str.strip().value_counts().to_dict())
```

Expected file names include the Monday, Tuesday, Wednesday, Thursday Web Attacks, Thursday Infiltration, Friday Botnet, Friday PortScan, and Friday DDoS files. The actual local file list and label counts settle any mirror-specific differences.

12.2 Evaluation split

Split	Source	Role
Train	Monday–Wednesday files	Model fitting and training-only preprocessing
Validation	Thursday files	Early stopping, threshold selection, hyperparameter decisions
Test	Friday files	Final held-out evaluation

Leakage rule: Never fit normalization, threshold, feature-selection, or other test-dependent choices on Friday data. Save exact file lists and hashes.

12.3 Dataset limitations

- The dataset is a benchmark environment, not a complete modern critical-infrastructure network.
- Raw labels do not equal ATT&CK tactics; the six-state mapping is project-defined.
- The small packet-derived layer is not a full reprocessing of every PCAP packet in the dataset.
- Host-specific patterns can create leakage-like memorization; track host overlap and scenario structure.
- Rare classes such as Heartbleed may not support stable statistical claims.

13. The three-tier research design

Tier 1 — Static classifier

$S_t \rightarrow \text{Logistic Regression} \rightarrow P(\text{attack at } t)$

Purpose: establish the performance of a non-temporal baseline. It answers “can the current window itself distinguish attack from benign behaviour?”

Tier 2 — Discrete Markov world model

$Z_t \rightarrow P(Z_{t+1} | Z_t) \rightarrow P(Z_{t+k} | Z_t)$

Purpose: operationalize the most literal state-transition form of the NTRO requirement with a transparent transition matrix. For six states, P is 6×6 and K-step forecasting can use matrix powers.

$P_{ij} = N(Z_t=i, Z_{t+1}=j) / N(Z_t=i)$

$p_{t+k} = p_t P^k$

Tier 3 — Attention-LSTM

$H_t = [S_{t-19}, \dots, S_t] \rightarrow \text{LSTM} \rightarrow \text{temporal attention} \rightarrow \text{forecast heads}$

Purpose: learn history-dependent dynamics that a first-order Markov table cannot express. The LSTM sees the full continuous state vector and can use multiple prior windows when the next outcome depends on recent trajectory rather than only the current discrete state.

14. Exact Attention-LSTM design

Parameter	Value
Input size	24
Sequence length	20
LSTM layers	1
Hidden size	64
Dropout	0.20 before forecast heads
Attention	Additive / Bahdanau-style over 20 hidden states
Risk output	Sigmoid
Tactic output	Softmax over 6 classes
Optimizer	Adam
Learning rate	1e-3
Batch size	64
Initial loss	weighted BCE + weighted CE across 5 horizons
Early stopping	Thursday validation loss

14.1 Additive attention

$e_i = v^T \tanh(W_h h_i + b_h)$

$\alpha_i = \text{softmax}(e_i)$

$c = \sum_i \alpha_i h_i$

The α_i values provide temporal evidence about which history positions the model weighted more strongly. They should not be described as causal proof.

15. Multi-horizon forecast heads

The final Week-1 LSTM has exactly two supervised heads at each of five future horizons.

Head	Definition	Activation	Evaluation
Future Risk	Any attack label active for host at $t+k$	Sigmoid	Brier/ECE, macro metrics, lead time
Future Tactic	One of BENIGN/RECON/INITIAL_ACCESS/LATERAL_MOVEMENT/C2/IMPACT at $t+k$	Softmax	Per-horizon macro-F1, horizon average

15.1 Direct multi-horizon versus recursive rollout

The Week-1 implementation uses direct outputs for $t+1\dots t+5$. This avoids feeding model predictions back into itself and accumulating error. Recursive rollout remains a Phase-2 experiment.

16. Markov versus LSTM: making the benchmark meaningful

The comparison must be aligned to the same conceptual forecasting question. The Markov model predicts the next discrete state from the current discrete state. The LSTM predicts future risk and tactic from a rich history of S_t vectors. The Markov model is intentionally simpler; the research question is whether the richer history-conditioned representation produces a measurable advantage.

16.1 Metric aggregation

$F1_k$ = macro-F1 at horizon k

$F1_K = (1/K) \sum_{k=1}^K F1_k$

Report $F1_1$ through $F1_5$ separately, then $F1_K$. The degradation curve is itself a result and should not be hidden behind a single average.

17. Success criteria fixed before seeing test results

1. Detection: Attention-LSTM improves macro-F1 over Logistic Regression by at least 5 percentage points, with FPR increase no greater than 2 percentage points.
2. Forecasting: Attention-LSTM improves horizon-averaged tactic macro-F1 over Markov by at least 5 percentage points; all five horizon scores are reported separately.
3. Lead time: median forecast lead time is at least one 10-second window ahead of labelled attack onset on at least 60% of test attack instances.

Why pre-register these bars: The team must not move the goalposts after seeing Friday results. A failure is a research result that tells us the present representation/model does not support the desired forecast quality.

18. Forecast lead time

$lead_time = attack_onset_time - first_valid_forecast_time$

The validation set determines the risk threshold and any persistence rule. That choice is frozen before test evaluation. Positive lead time means the forecast became valid before attack onset. Lead time near zero or negative means the system is behaving like a detector/reactor rather than a forecaster.

19. Calibration

Forecast probabilities should be treated as probabilities, not just scores. The team should evaluate Brier score and Expected Calibration Error (ECE) for the risk head.

$$\text{Brier} = \text{mean}((p_i - y_i)^2)$$

A “70% risk” output should be interpreted as a calibrated probability only if the empirical outcomes support that interpretation. Calibration is especially important for a security operator deciding whether to investigate or intervene.

20. Explainability

Two complementary signals are planned: SHAP for feature-level attribution and temporal attention for historical-window evidence.

- SHAP: Which input features are influential?
- Attention: Which historical windows receive more model weight?
- Use language such as “influential evidence” or “model-weighted history,” not “causal cause.”

21. ATT&CK integration

MITRE ATT&CK should be treated as semantic context after the numerical forecast. The model forecasts a project-defined behavioural class; the mapping layer then provides the relevant ATT&CK tactic/technique context. ATT&CK is not the probability engine and is not the native ground truth of CIC-IDS2017.

21.1 Terminology rule

Preferred: predicted ATT&CK tactic / likely technique

Avoid: “ATT&CK stage is guaranteed to occur next”

22. Vulnerability context — how it fits

The phrase “explore network vulnerability” describes an exposure-management problem, not the exact core forecasting task. Vulnerability/exposure data can become conditioning context in Phase 2: traffic tells us what is evolving, while exposure context tells us which assets are potentially reachable or consequential.

Future extension: $P(\text{future attack behaviour} \mid \text{observed telemetry, exposure/vulnerability context})$

For Week 1, do not make vulnerability scanning the core pipeline. It can distract from the main hypothesis: whether temporal network-state modelling improves forecast quality and lead time.

23. Why next-asset prediction is deferred

Predicting which specific destination asset will be contacted next is naturally a temporal link-prediction problem. A per-host vector model can approximate this only weakly because it discards explicit pairwise topology. Deferring it until a TGN/temporal-graph architecture is available is therefore an architectural strengthening, not an omission to hide.

24. Local AI agent — what it is and is not

The agent is the operator layer. It does not produce the core probability or forecast. It receives structured model output, retrieves context, calls bounded tools, summarizes evidence, and supports a human decision.

Forecast model → structured JSON → local agent → investigation / ATT&CK lookup / what-if → human decision

24.1 Minimum local agent design

Component	Week-1 choice
Runtime	Ollama
Model size target	3B–4B quantized; 7B–8B optional
Orchestration	Plain Python first
Cloud dependency	None
Numerical authority	Forecasting model only
Agent tools	host history, recent flows, ATT&CK lookup, explain forecast, what-if simulator, report generator
Framework	LangGraph optional later

24.2 Agent safety boundary

- Never invent or change probabilities.
- Never claim causal certainty from attention/SHAP.
- Do not autonomously isolate systems or delete accounts in the prototype.
- What-if numerical outcomes come from the model/simulator, not the LLM.
- Keep the knowledge base local for the offline story.

25. What-if simulation

A useful extension is to compare forecasts under a hypothetical state modification, such as suppressing a high-risk relation. The simulator compares the numerical forecast under H_t and H'_t .

Baseline: $H_t \rightarrow \text{risk curve}$

Scenario: $H'_t \rightarrow \text{risk curve}$

Compare: $\Delta \text{risk}(t+k)$

This is scenario analysis, not proof that a real intervention will succeed.

26. Why graphs and Mamba remain Phase 2

Temporal graphs naturally encode entities and relationships; Mamba/SSM offers efficient long-context sequence modelling. Both are valuable research directions found in the earlier gap analysis. They are

deliberately deferred because the one-week objective is a fully understood, evaluated prototype. Adding them without time for validation would turn the project into an architecture demo rather than a credible experiment.

27. Phase-2 research roadmap

1. Construct temporal graphs with host/entity nodes and time-stamped communication edges.
2. Use TGN/TGNN to learn relational embeddings.
3. Compare GRU/LSTM dynamics with Mamba/SSM.
4. Add next attacker–asset interaction as a link-prediction head.
5. Condition forecasts on network exposure/vulnerability context.
6. Evaluate cross-dataset and unseen-attack generalization.
7. Explore counterfactual/causal state-transition modelling only when a causal design is justified.
8. Study uncertainty, abstention, calibration, and operational alert policies.

28. Threats to validity

Threat	Effect	Mitigation
Temporal leakage	Unrealistically high scores	Chronological/scenario-aware split; train-only preprocessing
Host memorization	Model learns identity rather than behaviour	Track host overlap; inspect scenario structure
Label abstraction	Six buckets may not equal true tactics	Version mapping; clearly state it is project-defined
Rare classes	Unstable per-class metrics	Per-class support counts; avoid headline claims on tiny classes
Window mismatch	Important behaviour may straddle windows	Audit event density; freeze one design before test
Forecast imbalance	Model predicts frequent BENIGN too often	Weighted loss; macro-F1; per-class recall
Attention overinterpretation	Readers infer causality	Call it evidence, not cause
Agent hallucination	Narrative may diverge from model	Structured outputs and bounded tools
Benchmark mismatch	Different models answer different tasks	Define detection and forecasting comparisons separately

29. The complete algorithmic chain

1. Observe O_t
2. Aggregate into 24-D state S_t
3. Resolve raw labels into Z_t using precedence
4. Train static Logistic Regression baseline
5. Train Markov $P(Z_{t+1} | Z_t)$
6. Train Attention-LSTM over H_t
7. Produce five-horizon risk/tactic forecasts
8. Compute lead time, $F1_K$, calibration
9. Map forecast to ATT&CK context
10. Produce SHAP + attention evidence

11. Pass structured output to bounded local agent
12. Present in offline Streamlit SOC

30. What success would mean

Success does not mean DigitalSpy has solved cyberattack prediction in general. It means the experiment shows that, under the chosen dataset and split, temporal modelling of network states yields a measurable advantage in future-risk/tactic forecasting and/or advance warning compared with simpler baselines. The result then justifies Phase 2 graph/state-space research.

31. What failure would mean

If the Attention-LSTM does not beat the baselines or cannot produce positive lead time, the team should investigate the label abstraction, windowing, horizon, feature sufficiency, class imbalance, and dataset realism. A negative result does not invalidate the mathematical problem; it may show that the chosen observations do not contain enough predictive information at the chosen horizon.

32. Team Q&A — non-negotiable answers

Q: What exactly is your world model?

A: Tier 2 is a literal discrete state-transition model $P(Z_{t+1} | Z_t)$; Tier 3 is a learned temporal dynamics model over continuous S_t history that forecasts future risk and tactic distributions.

Q: Can you predict an attacker exactly?

A: No. We estimate probabilities of plausible future malicious behaviour from observed history.

Q: Why not GNN/Mamba?

A: They are Phase-2 extensions. The one-week prototype uses the permitted feature-vector representation so we can finish and evaluate a complete system.

Q: Why Markov?

A: It is the simplest transparent implementation of state transition and gives a strong sanity-check baseline.

Q: Why attention?

A: To provide inspectable temporal evidence about which history windows received more model weight.

Q: Is ATT&CK your ground truth?

A: No. We derive project-defined buckets from raw dataset labels; ATT&CK adds semantic context after prediction.

Q: Why not next target?

A: It is a graph/link-prediction problem and fits the Phase-2 temporal-graph architecture better.

Q: Why forecast lead time?

A: A forecast is operationally useful only if it gives a defender information before the event it predicts.

Q: Does the LLM make the prediction?

A: No. The forecasting model owns numerical predictions; the local agent interprets and orchestrates approved tools.

33. Reference set

[1] Canadian Institute for Cybersecurity, CIC-IDS2017 official dataset page:

<https://www.unb.ca/cic/datasets/ids-2017.html>

[2] Sharafaldin, Lashkari, Ghorbani, "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization," ICISSP 2018.

[3] Rossi et al., "Temporal Graph Networks for Deep Learning on Dynamic Graphs," arXiv:2006.10637. <https://arxiv.org/abs/2006.10637>

[4] Gu & Dao, "Mamba: Linear-Time Sequence Modeling with Selective State Spaces," arXiv:2312.00752. <https://arxiv.org/abs/2312.00752>

[5] "PARD-SSM: Probabilistic Cyber-Attack Regime Detection via Variational Switching State-Space Models," arXiv:2604.02299. <https://arxiv.org/abs/2604.02299>

[6] MITRE ATT&CK knowledge base and design/philosophy: <https://attack.mitre.org/>

[7] Internal research basis: "Competitive Technical Gap Analysis of Predictive Cyber Defence, Attack-Path Modelling, and Temporal Network Attack Forecasting for Critical Information Infrastructure in India."

34. Source and evidence notes

The team's uploaded research and prior internal documents were used as the main basis for the landscape, architectural alternatives, dataset roles, and implementation decisions. The current paper intentionally distinguishes documented source claims from project choices. The exact 24-feature schema, label precedence, windowing, split, success bars, and Week-1 boundaries are engineering decisions for this project, not claims that the source datasets or standards define them.

35. Final one-page mental model

Problem: defenders see present/past evidence while attacks evolve.

State: S_t = 24-D network-behaviour vector.

Transparent dynamics: $P(Z_{t+1} | Z_t)$.

Learned dynamics: Attention-LSTM over H_t .

Forecast: 5 future horizons for risk + tactic.

Evidence: SHAP + temporal attention.

Context: ATT&CK mapping.

Operator layer: local Ollama agent.

Success: measurable forecasting advantage + positive lead time.

Next phase: temporal graphs + Mamba + next-target prediction.